BoxDrive v3.26 Stepper Controller

Closed-Loop Motion Controller for NEMA 17 Stepper Motors

DS-BD326-REV326
Rev 3.26 • October 2026

01. ELECTRICAL & OPERATING SPECIFICATIONS

POWER & OPERATING ENVELOPE

PARAMETER	SPECIFICATION / VALUE	ENGINEERING DESCRIPTION & USAGE NOTES
Operating Supply Voltage	`9.0V – 24.0V DC`	Verified continuous operating voltage. Absolute maximum limit is 26.0V DC.
Continuous Motor Phase Current	`Up to 2.0A RMS`	Programmable via UART. Current above 1.2A RMS requires included aluminum heatsink.
Peak Instantaneous Current	`2.8A Peak`	Maximum transient acceleration current (duration < 10 ms).
Logic Voltage Level	`3.3V DC`	On-board DC-DC buck converter powers logic, ESP32-C6, and sensors from main DC bus.
Step Pulse Frequency	`Up to 500 kHz`	Supports high-speed step execution pulses with 1 µs minimum pulse width.
Operating Temperature	`-20°C to +70°C`	Verified operating temperature range: -20°C to +70°C ambient environment.

02. CORE HARDWARE & INTEGRATED SILICON

PRIMARY CHIPS & ROLES

COMPONENT	PART NUMBER	INTEGRATED FUNCTION & CAPABILITIES
Microcontroller (MCU)	`ESP32-C6FH4`	32-bit RISC-V @ 160 MHz, 4MB Flash, Wi-Fi 6 (802.11ax), and Bluetooth 5.0 (BLE).
Stepper Driver IC	`Trinamic TMC2209-LA`	Ultra-silent StealthChop2™, SpreadCycle™ torque mode, StallGuard4™ sensorless homing.
Magnetic Angle Encoder	`AS5600-ASOM`	12-bit contactless on-axis Hall sensor (4096 counts/rev, 0.0879° resolution) over I2C.
CAN Bus Transceiver	`SN65HVD230D`	3.3V CAN 2.0B differential transceiver (up to 1.0 Mbps) with ±16V fault tolerance.

03. PINOUT & GPIO ASSIGNMENTS

COMPLETE TERMINAL CONNECTION TABLE

PORT / PIN	SIGNAL	TYPE	TERMINAL FUNCTION & CONNECTION DESCRIPTION
`GPIO 0`	`TMC_DIR`	Output	Direction control: HIGH = Clockwise (CW), LOW = Counter-Clockwise (CCW).
`GPIO 1`	`TMC_STEP`	Output	Step execution clock pulse (hardware timer driven up to 500 kHz).
`GPIO 2`	`TMC_EN`	Output	**Active LOW:** LOW = Coils energized (running), HIGH = Motor free (freewheel).
`GPIO 4`	`USER_IO`	I/O (Free)	**Free User Pin:** Broken out on board for custom application use (Digital I/O, PWM servo, endstop/limit switch, or interrupt).
`GPIO 14`	`TMC_RX`	Input	TMC2209 UART RX single-wire line for driver telemetry and register readback.
`GPIO 18`	`TMC_TX`	Output	TMC2209 UART TX single-wire line for current scaling and chopper tuning.
`GPIO 19`	`STATUS_LED`	Output	Active-HIGH green SMD status LED for heartbeat and error blink codes.
`GPIO 22`	`I2C_SCL`	OD	I2C serial clock for AS5600 angle sensor (on-board 4.7 kΩ pull-up).
`GPIO 23`	`I2C_SDA`	OD	I2C serial data for AS5600 angle sensor (on-board 4.7 kΩ pull-up).
`J1, J3`	`XT30PW (2+2)`	Pwr / CAN	Dual pass-through ports: DC power (9.0V–24.0V, 15A rated) + Differential CAN bus.
`J2`	`4-Pin 2.54mm`	Coils Out	Bipolar stepper motor coil phases: AA, AB (Coil A) and BA, BB (Coil B).
`J4`	`USB Type-C`	USB 2.0	Native ESP32-C6 USB-CDC / JTAG for direct flashing and serial CLI monitor.

04. MECHANICAL DIMENSIONS & CAD ENVELOPE

DIRECT NEMA 17 FLANGE MOUNT

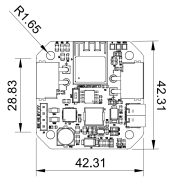


FIGURE 1: TOP ENVELOPE (42.31 MM, 28.83 MM PITCH)

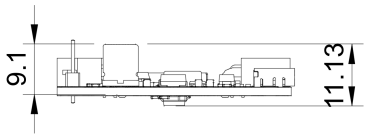


FIGURE 2: PROFILE (9.10 MM TOP, 1.13 MM SENSOR)

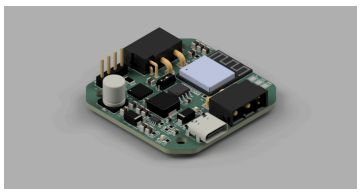


FIGURE 3: POPULATED HARDWARE ASSEMBLY

MECHANICAL PARAMETER	NOMINAL VALUE	ENGINEERING DESCRIPTION & TOLERANCE
PCB Outer Dimensions	`42.31 × 42.31 mm`	±0.10 mm tolerance. Matches standard NEMA 17 rear motor flange.
Mounting Hole Pitch	`28.83 mm`	±0.10 mm tolerance. Center-to-center square pattern aligned with NEMA 17 holes.
Mounting Hole Specification	`4× ø3.30 mm CSK`	Countersunk clearance holes designed for flush-mounted M3 machine screws.
Top-Side Component Clearance	`9.10 mm`	Height to top apex of XT30PW connector and low-ESR bulk electrolytic capacitor.
Bottom-Side Sensor Clearance	`1.13 mm`	Height of AS5600 angle encoder IC above the bottom surface of the PCB.
Net Board Assembly Mass	`18.0 grams`	Net weight of populated controller board including aluminum laser cut heatsink.

05. CAN BUS 120Ω TERMINATION & DAISY-CHAIN ARCHITECTURE

INTEGRATED HARDWARE SLIDE SWITCH

On-Board 120 Ω Termination Slide Switch: SLIDE SWITCH

BoxDrive features an integrated slide switch on the PCB. Flipping the switch to **ON** directly enables the internal 120 Ω termination resistor across CAN_H and CAN_L, eliminating external soldered resistors.

MULTI-NODE DAISY-CHAIN CONFIGURATION EXAMPLE (4 NODES):

NODE A (Master / Start)	NODE B (Joint 2)	NODE C (Joint 3)	NODE D (Wrist / End)
120Ω Switch: ON	120Ω Switch: OFF	120Ω Switch: OFF	120Ω Switch: ON
Bus Endpoint 1	Intermediate Node	Intermediate Node	Bus Endpoint 2

Rule: Set the slide switch to **ON** exclusively at the two extreme physical ends of the daisy-chain bus. All intermediate pass-through nodes must have termination set to **OFF**.

Standalone Operation: If operating BoxDrive as a standalone single controller (USB serial, Wi-Fi WebUI, pulse/direction step-stick mode, or direct point-to-point without an extended daisy-chain bus), the termination switch position does not matter.

06. OPERATIONAL DIRECTIVES & SAFETY MANDATES

MANDATORY ENGINEERING RULES

CRITICAL SAFETY MANDATES:

- **Minimum Operating Voltage:** Input voltage must be at least **9.0V DC** (safe operating window is 9.0V to 24.0V DC). Exceeding 26.0V DC causes destructive overvoltage.
- **Inductive Back-EMF Protection:** Never unplug the J2 motor coil connector while DC power is connected. Breaking an energized motor circuit produces high-voltage inductive flyback spikes (>100V) that will destroy the TMC2209 MOSFET bridge.
- **Driver Enable Logic:** GPIO 2 (TMC_EN) is strictly **Active LOW** (pulling LOW energizes coils; pulling HIGH disables bridge into freewheel mode).

07. SOFTWARE LIBRARY & FIRMWARE CODE EXAMPLES

OFFICIAL GITHUB REPOSITORY

Official BoxDrive Arduino C++ Library (20+ Examples)
Verified code examples for CAN multi-axis synchronization, closed-loop position control, WebUI tuning, and StallGuard4.

`github.com/BoxPoxin/BoxDrive_Library →`